L5: Frequent Motifs and Their Function

Network	Nodes	Edges	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score	N_{real}	$N_{rand} \pm SD$	Z score
Gene regulation (transcription)					**Feed-forward loop**			**Bi-fan**			
E. coli	424	519	40	7 ± 3	10	203	47 ± 12	13			
*S. cerevisiae**	685	1,052	70	11 ± 4	14	1812	300 ± 40	41			
Neurons					**Feed-forward loop**			**Bi-fan**			**Bi-parallel**
C. elegans†	252	509	125	90 ± 10	3.7	127	55 ± 13	5.3	227	35 ± 10	20
Food webs					**Three chain**			**Bi-parallel**			
Little Rock	92	984	3219	3120 ± 50	2.1	7295	2220 ± 210	25			
Ythan	83	391	1182	1020 ± 20	7.2	1357	230 ± 50	23			
St. Martin	42	205	469	450 ± 10	NS	382	130 ± 20	12			
Chesapeake	31	67	80	82 ± 4	NS	26	5 ± 2	8			
Coachella	29	243	279	235 ± 12	3.6	181	80 ± 20	5			
Skipwith	25	189	184	150 ± 7	5.5	397	80 ± 25	13			
B. Brook	25	104	181	130 ± 7	7.4	267	30 ± 7	32			

Most real-world networks are either designed by humans *(such as electronic circuits or communication networks) or they evolve naturally (such as biological or social networks)* to perform certain functions.

Consequently, the frequent presence of a network motif in a network suggests that that specific connectivity pattern has a functional role in the network. For example, the frequent presence of the Feedback Loop motif *(see FBL motif in previous pages)* suggests that the network includes control loops that regulate the function of a three-node network path with feedback from the output node back to the input node.

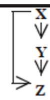
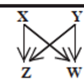
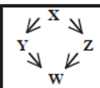
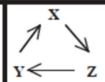
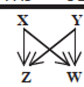
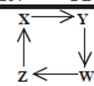
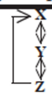
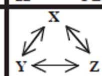
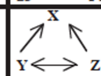
Similarly, the absence of a network motif from a network suggests that that connectivity pattern is functionally *"not allowed"* in that kind of network. For example, a hierarchical network that shows who is reporting to whom at a company should not contain any motifs that include a directed cycle.

The research paper that we refer to in this page has analyzed a large variety of networks (gene regulatory networks, neuronal networks, food webs in ecology, electronic circuits, the WWW, etc) and identified the network motifs that are most frequently seen in each type of network. Their study did not consider only 3-node motifs but also larger motifs. For each type of network, the authors identify a few common network motifs and associated a plausible function for that motif.

For instance, in gene regulatory networks the FeedForward Loop (FFL) motif is a prevailing structure.

In that context, an FFL instance consists of three genes *(X,Y,Z)*: two input transcription factors (*X* and *Y*), one of which regulates the other ($X \rightarrow Y$), both jointly regulating a target gene *Z*. The FFL has eight possible structural types, because each of the three interactions in the FFL can be activating or repressing. Four of the FFL types, termed incoherent FFLs, act as sign-sensitive accelerators: they speed up the response time of the target gene expression following stimulus steps in one direction (e.g., off to on) but not in the other direction (on to off). The other four types, coherent FFLs, act as sign-sensitive delays. For additional information about the biological function of the FFL motif, please see the following research paper:

Structure and function of the feed-forward loop network motif, S. Mangan and U. Alon, PNAS October 14, 2003 100 (21) 11980-11985 (https://doi.org/10.1073/pnas.2133841100)

Electronic circuits (forward logic chips)					**Feed-forward loop**			**Bi-fan**			**Bi-parallel**
s15850	10,383	14,240	424	2 ± 2	285	1040	1 ± 1	1200	480	2 ± 1	335
s38584	20,717	34,204	413	10 ± 3	120	1739	6 ± 2	800	711	9 ± 2	320
s38417	23,843	33,661	612	3 ± 2	400	2404	1 ± 1	2550	531	2 ± 2	340
s9234	5,844	8,197	211	2 ± 1	140	754	1 ± 1	1050	209	1 ± 1	200
s13207	8,651	11,831	403	2 ± 1	225	4445	1 ± 1	4950	264	2 ± 1	200
Electronic circuits (digital fractional multipliers)					**Three-node feedback loop**			**Bi-fan**			**Four-node feedback loop**
s208	122	189	10	1 ± 1	9	4	1 ± 1	3.8	5	1 ± 1	5
s420	252	399	20	1 ± 1	18	10	1 ± 1	10	11	1 ± 1	11
s838‡	512	819	40	1 ± 1	38	22	1 ± 1	20	23	1 ± 1	25
World Wide Web					**Feedback with two mutual dyads**			**Fully connected triad**			**Uplinked mutual dyad**
nd.edu§	325,729	1.46e6	1.1e5	2e3 ± 1e2	800	6.8e6	5e4±4e2	15,000	1.2e6	1e4 ± 2e2	5000

In food webs, on the other hand, we rarely see the FFL motif. The reason is that if a carnivore species *X* eats a herbivore species *Y*, and *Y* eats a plant species *Z*, we rarely see that *X* also eats *Z*.